

SIVAKAMI SRINIVASAN

Candidate for Data Analytics / Business Intelligence / Operations Analytics Internship (Winter 2027)

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[My Tableau Portfolio](#)

PROFESSIONAL SUMMARY

Master of Data Analytics candidate (GPA: 3.78) seeking a Winter 2027 Internship in Data Analytics, Operations Research, or BI. Combines 8+ years of engineering experience in software debugging and system optimization with expertise in statistical modeling (Regression, Classification, ANOVA, ARIMA/SARIMAX), mathematical optimization (PuLP, OR-Tools), Python, SQL, and Tableau. Proven track record in executing rigorous hypothesis testing, forecasting dynamic risk trends, and converting complex optimization and statistical outputs into executive decision dashboards.

TECHNICAL SKILLS

- **Statistical Modeling & Analytics:** Regression Analysis (Linear/OLS, Logistic), Classification Models, Hypothesis Testing (ANOVA, Mann-Whitney U, Welch's t-test), Time Series Forecasting (ARIMA, SARIMAX), Statistical Significance Testing
- **Optimization & Prescriptive Analytics:** Integer Linear Programming (ILP), Ant Colony Optimization (ACO), Linear Programming, Decision Trees, Monte Carlo Simulation, Data Envelopment Analysis (DEA), Supply Chain & Operations Analytics
- **Languages & Scripting:** Python (Pandas, NumPy, Scikit-Learn, Statsmodels, PuLP, OR-Tools, XGBoost, LightGBM, Matplotlib), SQL (MySQL), R, Bash, C/C++
- **Data Visualization & BI:** Tableau Desktop (Calculated Fields, Parameters, Geospatial Mapping, Dashboard Actions), Power BI (DAX, Data Modeling), Advanced Excel (PivotTables, Statistical Functions)
- **Data Engineering & Cloud:** AWS (S3, Redshift, Glue, Athena, Lambda), ETL Pipelines, Relational Database Schema Design (DDL, DML, DQL), Data Munging

GRADUATE ANALYTICS PROJECTS

Mortgage Underwriting Risk Assessment & Capital Optimization | Python (PuLP, XGBoost, LightGBM, Statsmodels), MySQL, Tableau

- **Hypothesis Testing & Statistical Modeling:** Engineered Logistic Regression, XGBoost, and LightGBM classification models to evaluate loan default indicators; performed rigorous hypothesis testing confirming DTI and CLTV financial risk ratios as statistically significant drivers of mortgage denials in Virginia.
- **Prescriptive Capital Allocation:** Formulated an Integer Linear Programming (ILP) model in Python (PuLP) that optimized portfolio return while strictly enforcing a 3.5% aggregate denial-risk cap against static rule benchmarks.
- **Executive BI Dashboards:** Designed interactive Tableau dashboards showcasing portfolio return trade-offs, risk boundaries, and predictive discrimination thresholds for executive decision-making.

AI Industry Impact Analysis & PESTEL Strategic Framework | Python, Statsmodels, Scikit-Learn, Machine Learning

- **Time Series Forecasting (ARIMA/SARIMAX):** Engineered SARIMAX / ARIMA time series models to forecast monthly financial risk exposure and adoption timelines across 157,000+ patent records and cybersecurity incident logs.
- **Classification & Clustering:** Applied Random Forest classification for cyber-loss pattern identification and K-Means Clustering on innovation fields to segment key industry sectors and predict patent approval timelines.

Population Health Diabetes Classification & BMI OLS Regression Pipeline | Python, Statsmodels, Scikit-Learn

- **OLS Regression & ANOVA Testing:** Developed an Ordinary Least Squares (OLS) regression pipeline on population survey data, performing ANOVA-based feature contribution analysis, log-transformations for heteroscedasticity, and VIF checks for multicollinearity.
- **Classification ML Pipeline:** Built a classification pipeline on 456K+ CDC records using SMOTE oversampling and Random Forest ensemble models, achieving an ROC-AUC of 0.795 in predicting diabetes diagnoses.

Stochastic Supply Chain & Operations Routing Optimization | Python, OR-Tools, MySQL

- **Operations Research Extension:** Extended peer-reviewed operations research literature (Wiley 2019) in Python (OR-Tools) to incorporate stochastic weather variables, optimizing routing schedules under dynamic risk constraints.

RELEVANT EXPERIENCE

Firmware Engineer (Contract) | AMD Canada, Markham, ON Sep 2024 – Nov 2025

- **Data Analysis & Calibration:** Executed numerical data processing and register calculations (CDD/ODT) to calibrate DDR5 PHY memory parameters on RISC-V architecture.

Senior Software Engineer | Qualcomm, India Jan 2019 – May 2023

- **Log Analytics & System Performance:** Parsed multi-gigabyte system execution logs to diagnose root-cause anomalies for enterprise clients (Cisco, Aruba), optimizing system performance metrics.

Software Engineer / Senior Software Engineer | Visteon & Tata Elxsi, India Jul 2012 – Jun 2017

- **Software & Data Transformation:** Engineered C++/Java graphics modules and implemented data transformation logic (RGB models, alpha blending) for automotive displays.

EDUCATION & CERTIFICATIONS

Master of Data Analytics | University of Niagara Falls, Canada (GPA: 3.78 / 4.33) Expected Graduation: 2027

Bachelor of Engineering in Electronics & Communication | Rajalakshmi Engineering College, India (CGPA: 8.87 / 10.0)

- **Certifications:** AWS Certified Data Engineer – Associate (Candidate, Expected Dec 2026) | Google Data Analytics Professional Certificate (2025)